



# **AeroTHON** **2026**

DESIGN REPORT  
TRACK 1: ROTORCRAFT SYSTEMS CHALLENGE

**CLUB AEROUNWIRED**  
NATIONAL INSTITUTE OF TECHNOLOGY, CALICUT

TEAM ID : AT2026RC11



## **TEAM MEMBERS**

PARVATHY G  
SRINAND MANKALA  
DILAN A L  
SHREYA N SHENOY  
NAVANEETH P  
ANIRUDH S  
MOHAMED SHAHID VILLAN  
SHIVAM VISHWAJIT KAKADE  
RISHIRAJ PARIDA  
ADITHYAN C S

## APPENDIX A

### STATEMENT OF COMPLIANCE

#### Certification of Qualification

Team Name: CLUB AEROUNWIRED

University/Institute: NATIONAL INSTITUTE OF TECHNOLOGY CALICUT

Faculty Advisor: Vijayaraj K

Faculty Advisor's Email: vijayaraj@nitc.ac.in

#### Statement of Compliance:

As Faculty Advisor, I certify that the registered team members are enrolled in collegiate courses. This team has designed the UAS for the SAE AEROTHON 2026 contest, without direct assistance from professional engineers, R/C model experts or pilots, or related professionals.



Signature of Faculty Advisor

24/05/2026

Date

#### Team Captain Information:

Team Captain's Name: PARVATHY G

Team Captain's E-mail: paru2004parvathy@gmail.com

Team Captain's Phone: 7356090316

#### **Note:**

*A copy of this statement needs to be included in your Design Report as page 2*

# Contents

|          |                                                       |           |
|----------|-------------------------------------------------------|-----------|
| <b>1</b> | <b>Conceptual Design</b>                              | <b>5</b>  |
| 1.1      | System Overview . . . . .                             | 5         |
| 1.2      | Mission Flow . . . . .                                | 7         |
| <b>2</b> | <b>Detailed Design</b>                                | <b>8</b>  |
| 2.1      | Thrust Requirements Estimation . . . . .              | 8         |
| 2.2      | Propulsion System Selection . . . . .                 | 9         |
| 2.3      | UAS Performance Estimation . . . . .                  | 10        |
| 2.4      | Material Selection . . . . .                          | 13        |
| 2.5      | Subsystem Selection . . . . .                         | 14        |
| 2.6      | Stability Analysis of the Quadcopter . . . . .        | 18        |
| 2.7      | Frame Design and Analysis . . . . .                   | 20        |
| 2.8      | Computational Analysis . . . . .                      | 20        |
| 2.9      | Payload Mechanism . . . . .                           | 23        |
| 2.10     | Safety Mechanisms . . . . .                           | 23        |
| 2.11     | Detailed Weight Breakdown and C.G. Analysis . . . . . | 24        |
| 2.12     | UAS sizing . . . . .                                  | 25        |
| <b>3</b> | <b>Methodology for Autonomous Operations</b>          | <b>26</b> |
| 3.1      | QR Code Detection and Identification . . . . .        | 26        |
| 3.2      | Delivery Zone Identification . . . . .                | 27        |
| 3.3      | Restricted Zone Avoidance . . . . .                   | 27        |
| 3.4      | Payload Delivery Methodology . . . . .                | 27        |
| <b>4</b> | <b>Final UAS Specifications and Bill of Materials</b> | <b>29</b> |
| <b>5</b> | <b>Innovation and Conclusion</b>                      | <b>30</b> |
| 5.1      | Innovation & System Optimization . . . . .            | 30        |
| 5.2      | Conclusion . . . . .                                  | 30        |
| <b>A</b> | <b>Appendix</b>                                       | <b>31</b> |
| A.1      | Datasheets . . . . .                                  | 31        |
| A.2      | Full Assembly View . . . . .                          | 33        |
| A.3      | Plansheet . . . . .                                   | 34        |
| A.4      | Part Drawing . . . . .                                | 35        |
|          | <b>References</b>                                     | <b>36</b> |

## **Abstract**

The goal of this project, which was created for the AeroTHON 2026 Rotorcraft Systems Challenge, is to design and develop a lightweight, stable, and efficient autonomous drone capable of performing precision payload delivery, obstacle navigation, QR-based autonomous operations, and safe return-to-home functionality within the competition constraints. The design process encompasses comprehensive trade studies, structural analysis, aerodynamic evaluation, and autonomous system integration to meet both manual and autonomous mission requirements. This report details the conceptual design approach, component selection methodology, structural and computational analysis, autonomous navigation algorithms, and mission execution strategy. The final design demonstrates a well-integrated multirotor system optimized for reliable obstacle avoidance, accurate observation capabilities, QR code detection and navigation, restricted zone awareness, and precision payload deployment. Performance analysis indicates the system meets all competition specifications while maintaining appropriate safety margins and operational efficiency across both flight mission.

## **Introduction**

This report presents the final competition design for an autonomous drone system capable of performing both manual and fully autonomous mission scenarios. The AeroTHON 2026 Rotorcraft Systems Challenge requires participating teams to design, build, and demonstrate an Uncrewed Aircraft System (UAS) that can execute complex flight operations including manual obstacle navigation with observation tasks, and fully autonomous QR-based navigation with precision payload delivery.

The competition establishes stringent performance requirements across two distinct flight missions. Mission 1 demands skilled manual piloting through a structured obstacle course while maintaining stable flight control, avoiding restricted zones, performing aerial observation tasks, and delivering payload to a designated target. Mission 2 requires complete autonomous operation, where the UAS must scan QR codes to determine delivery locations, navigate through confined corridors, identify correct delivery zones while avoiding restricted areas, and execute precision payload deployment before returning safely to the launch point. Meeting these objectives necessitates a multidisciplinary engineering approach integrating airframe design, propulsion system optimization, autonomous navigation algorithms, sensor integration, and robust control systems. This report documents the complete design process, from initial conceptual development through detailed analysis and final system integration, demonstrating how engineering principles and computational tools were applied to develop a competition-ready autonomous rotorcraft system.

# 1. Conceptual Design

## 1.1 System Overview

The UAS is designed as a lightweight autonomous quadrotor optimised for the AeroTHON 2026 mission profile. The platform integrates a custom carbon fibre frame with commercial off-the-shelf (COTS) avionics, a dedicated compute stack, and a precision payload delivery mechanism. The overall system is organised into five primary subsystems as described below.

### Airframe

The airframe is a custom-designed carbon fibre quadrotor frame selected for its high stiffness-to-weight ratio. The layout follows a symmetric X-configuration to ensure uniform thrust distribution and predictable flight dynamics. All electronics and subsystems are mounted centrally to maintain a low centre of gravity.

### Propulsion System

Table 1.1: Propulsion System Specifications

| Sl. No. | Component          | Specification                 |
|---------|--------------------|-------------------------------|
| 1       | Motor              | 1050 KV brushless, $\times 4$ |
| 2       | Propeller          | 8 inch tri-blade              |
| 3       | Battery            | 5000 mAh Lithium-Ion          |
| 4       | Target Flight Time | $\geq 15$ minutes             |

The propulsion system employs four 1050 KV brushless motors paired with 8-inch tri-blade propellers, selected to balance thrust output and power efficiency within the 2 kg MTOW constraint. A 5000 mAh lithium-ion battery pack provides the energy required to achieve a target endurance of at least 15 minutes under nominal mission loading.

### Avionics and Compute Stack

Table 1.2: Avionics and Compute Components

| Sl. No. | Component                     | Function                                      |
|---------|-------------------------------|-----------------------------------------------|
| 1       | Orange Cube flight controller | Stabilisation, attitude control, actuation    |
| 2       | Jetson Nano                   | Mission compute, sensor fusion, AI inference  |
| 3       | Stereo camera (forward)       | Obstacle detection, entry/exit identification |
| 4       | ArduCam (downward)            | QR code scanning, red zone detection          |
| 5       | Range Sensor(downward)        | Altitude hold and terrain following           |

The avionics stack is split into two layers. The Orange Cube flight controller handles all low-level stabilisation and motor actuation via ArduPilot firmware. The Jetson Nano serves as the high-level mission computer, running ROS to process sensor data and dispatch navigation commands to the Orange Cube over a serial MAVLink interface. Perception is handled by a forward-facing stereo camera for obstacle and passage detection, a downward-facing ArduCam for QR code scanning and red zone identification, and a downward LiDAR for precise altitude hold.

## Payload Delivery Mechanism

The payload mechanism is designed to deliver a 100 g package (10 cm × 5 cm × 5 cm) while the UAS maintains a stationary hover. Rather than a drop release, the mechanism lowers the payload gradually via a motorised descent system, ensuring controlled and precise placement at the target point with minimal disturbance to the flight attitude.

## System Architecture

Figure 1.1 illustrates the high-level data and command flow between all primary subsystems.

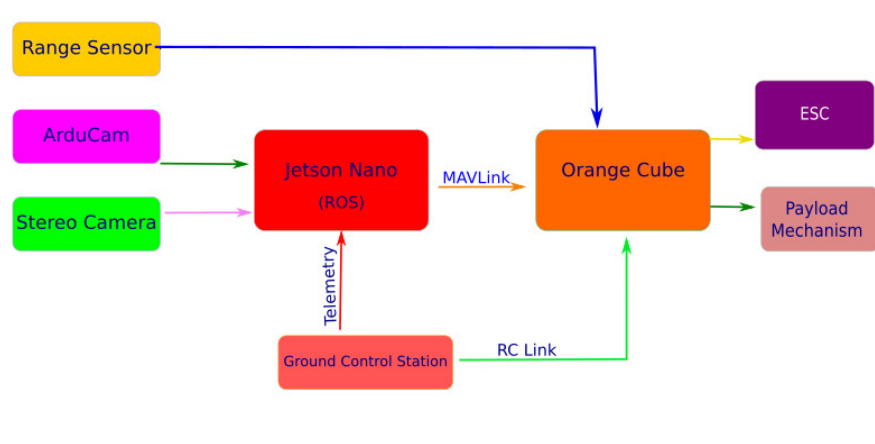


Figure 1.1: High-Level System Architecture

## Mission Requirements

### Manual Mission

The drone must be capable of manually navigating through a series of obstacles with sufficient flight endurance, an effective FPV system, and an adequate thrust-to-weight ratio to ensure stable and responsive piloting.

### Autonomous Mission

The autonomous mission requires the UAS to complete a structured sequence of tasks without manual intervention. The drone first scans an entry QR code, navigates through the arena passage, and ascends to a survey altitude of 10 m. It then systematically surveys the arena interior, avoiding walls and red zones, while scanning QR codes until a match with the entry code is found. Upon confirmation, the drone locates the exit and egresses the arena autonomously.

## 1.2 Mission Flow

Figure 1.2 provides a high-level overview of the autonomous mission sequence executed by the UAS.

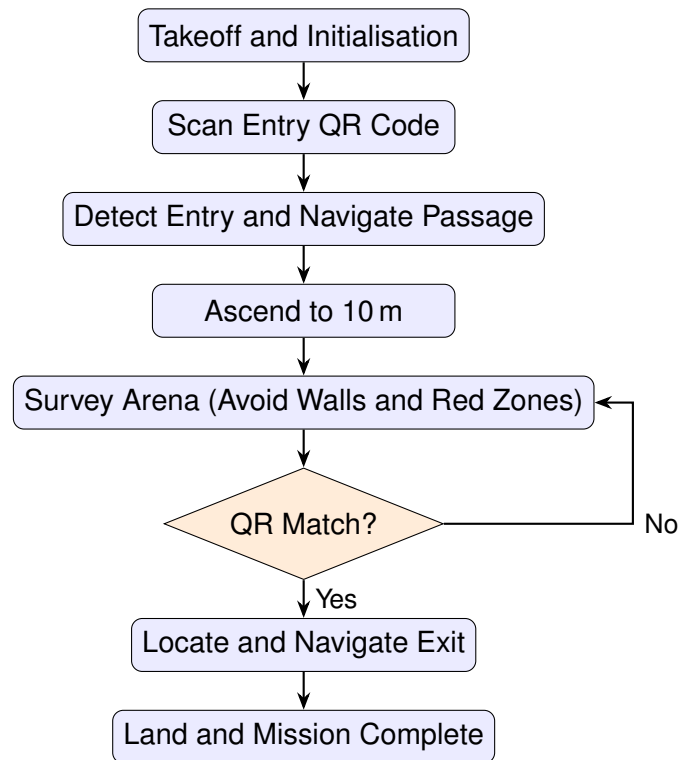


Figure 1.2: Autonomous Mission Flow

## 2. Detailed Design

### 2.1 Thrust Requirements Estimation

#### Total Thrust Calculation

The drone is estimated to have an all-up weight of 2 kg, including the frame, battery, propulsion system, avionics, and payload components.

Thus, the total downward gravitational force acting on the drone is calculated as:

$$w = mg = 2 \text{ kg} \times 9.81 \text{ m/s}^2 = 19.62 \text{ N}$$

Therefore, the total weight acting on the drone is:

$$19.62 \text{ N}$$

#### Thrust-to-Weight Ratio

For stable maneuverability and safe operation, drones generally maintain a thrust-to-weight ratio of approximately 2 : 1.

The selected design uses a target thrust-to-weight ratio of:

$$T : W = 2 : 1$$

Therefore, the required maximum thrust becomes:

$$T_{\max} = 2 \times 2000 \text{ g} = 4000 \text{ g}$$

Since the quadcopter uses four motors, the thrust required per motor is:

$$T_{\text{motor}} = \frac{4000}{4} = 1000 \text{ g}$$

Thus, each motor should ideally provide approximately 1 kg of thrust.

#### Hover Thrust Estimation

During hover, the total thrust required is equal to the total weight of the drone.

The thrust required per motor during hover is:

$$T_{\text{hover}} = \frac{2000 \text{ g}}{4} = 500 \text{ g per motor}$$

Based on interpolation from the motor datasheet, the SpeedyBee 2807 1050KV motor with the Gemfan 8040 3-blade propeller produces approximately 500 g of thrust at a current draw of around 3.4 A.

Therefore, the total propulsion current during hover becomes:

$$I_{\text{hover(propulsion)}} = 4 \times 3.4 \approx 13.6 \text{ A}$$

Including the avionics current consumption of approximately 1.8 A:

$$I_{hover(total)} = 13.6A + 1.8A \approx 15.4A$$

Therefore, the estimated total hover current of the UAV is approximately 15.4 A.

## Safety Factor Considerations

The propulsion system was selected such that the UAV maintains a thrust-to-weight ratio greater than 2:1 during normal operation. However, for safety, stability, and improved controllability, a throttle limit can be applied to restrict the maximum usable thrust.

A practical throttle limit was considered so that the effective thrust-to-weight ratio remains close to:

$$2.5 : 1$$

This provides sufficient thrust reserve for stable hovering, maneuvering, payload carrying, and recovery from wind disturbances, while also preventing excessive current draw and improving overall flight efficiency.

The additional thrust margin also reduces motor stress during aggressive throttle changes and helps maintain stable autonomous flight performance.

## 2.2 Propulsion System Selection

### Motor Selection

The **SpeedyBee 2807 1050KV (6S)** brushless motor was selected for its ability to generate high thrust efficiently while supporting medium-to-heavy payload configurations. Motor datasheet performance values are tested using 8-inch propellers, matching this design's propeller selection to ensure optimal efficiency and adequate thrust with controlled current draw.

Refer Appendix Figure A.2: 3D CAD model and Figure A.4: Motor datasheet efficiency.



Figure 2.1: SpeedyBee 2807 1050KV

### ESC Selection

The **GEP-BLS50A-4IN1 ESC** was selected for its reliable high-current performance in a compact, lightweight form. It supports 6S Li-Ion input and delivers 50 A continuous current per motor. DShot protocol support ensures precise low-latency throttle control, critical for stable autonomous flight. The 4-in-1 design reduces wiring complexity and overall weight, resulting in a cleaner and more efficient build.

Refer Appendix Figure A.3: ESC Specifications.

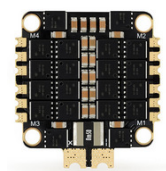


Figure 2.2: GEP-BLS50A-4IN1 ESC

### Propeller Selection

The **Gemfan 8040 CL Cinelifter 3-Blade** propeller was selected for its 8-inch diameter and 4-inch pitch, providing an excellent balance of thrust, efficiency, and stiffness. The 4-inch

pitch delivers strong forward bite without overloading the motor, enabling quick acceleration and stable performance across a wide throttle range.

## Battery Compatibility

The battery used is **iFlight Fullsend 6S 5000 mAh Li-Ion Battery**.  
Maximum theoretical current delivery:

$$I_{\max} = \text{Capacity} \times \text{C-rating} = 5 \times 25 = 125 \text{ A}$$

This significantly exceeds the expected maximum system current, ensuring adequate power availability during aggressive throttle conditions. Refer Appendix Figure A.1 for full battery specifications.



Figure 2.3: iFlight Fullsend 6S 5000 mAh Li-Ion

## Justification of Selected Propulsion System

The selected propulsion system provides sufficient thrust, stable flight performance, and good endurance for a UAV with an estimated takeoff weight of about 2 kg and provides a thrust-to-weight ratio greater than 2:1. The **SpeedyBee 2807 1050KV** and **Gemfan 8040 CL Cinelifter 3-Blade** was chosen because it provides good torque and stable thrust and supports 3-blade operation within safe current limits, making the 2807 1050KV and 8040 propeller an effective pairing for endurance missions

The **GEP-BLS50A-4IN1 ESC** was selected because it supports up to 50 A continuous current per motor, which is significantly higher than the expected operating current of 10A per motor for 1kg thrust, thus keeping a safe room of upto 40 A and have Burst Current 55A for 5 Seconds. This provides safe operation during aggressive throttle changes and reduces the possibility of overheating. The 4-in-1 design also reduces wiring complexity and overall system weight.

**iFlight Fullsend 6S 5000 mAh Li-Ion battery** was selected for its high energy density and lower weight. With a 25C discharge rating, the battery can theoretically deliver:

$$I_{\max} = 5 \times 25 = 125 \text{ A}$$

which is well above the expected maximum current requirement of 50 A, ensuring reliable power delivery during takeoff, manoeuvring, and autonomous flight.

Li-Ion was preferred over Lithium-Polymer (LiPo) for three reasons: the rigid metal casing and liquid electrolyte construction offers better protection and steady long-duration discharge; Li-Ion cells have a significantly longer cycle life; and at equivalent weight, Li-Ion provides higher capacity than LiPo, directly supporting the 15-minute endurance target.

## 2.3 UAS Performance Estimation

### Power Requirements

The total power consumption of the UAS consists of propulsion power and avionics power.

$$P_{\text{avionics}} \approx 35W$$

Assuming battery voltage:

$$V_{battery} = 22.2V$$

Equivalent avionics current:

$$I = \frac{P}{V}$$

$$I_{avionics} = \frac{35}{22.2} \approx 1.6A$$

Considering regulator losses and transient load variation:

$$I_{avionics(practical)} \approx 2A$$

Thus, the avionics stack continuously consumes approximately 2 A during operation.

The propulsion power during hover is estimated using the selected SpeedyBee 2807 1050KV motor with Gemfan 8040 3-blade propeller.

For a 2 kg quadrotor:

$$T_{hover(motor)} = \frac{2000}{4} = 500g$$

From the motor thrust table, approximately 500 g thrust requires:

$$I_{hover(motor)} \approx 3.4A$$

Therefore,

$$I_{hover(propulsion)} = 4 \times 3.4 \approx 13.6A$$

Including avionics current:

$$I_{hover(total)} = 13.6A + 2A \approx 15.6A$$

The corresponding hover power becomes:

$$P = VI$$

$$P_{hover} = 22.2 \times 15.6 \approx 346W$$

Hence, the estimated hover power requirement of the UAV is approximately 350 W.

## Battery Selection

The selected battery is the **I Flight Fullsend 6S 5000mAh Lithium-Ion Battery** with a nominal voltage of 22.2 V and a discharge rating of 25C.

The battery was selected mainly because of its high energy density and lower weight compared to conventional LiPo batteries. Even with onboard AI computing and sensing modules operating continuously, the battery provides acceptable flight endurance while keeping the total UAV weight manageable.

The 25C discharge rating allows the battery to safely provide high current during takeoff, maneuvering, and sudden throttle changes without significant voltage sag.

Maximum theoretical discharge current:

$$I_{max} = Capacity \times C\text{-rating} = 5 \times 25 = 125A$$

This is significantly higher than the expected operating current of the UAV.

The battery also weighs only approximately 436 g, which helps improve the overall thrust-to-weight ratio and endurance of the system.

## Battery Capacity Estimation

The total battery capacity is:

$$C_{battery} = 5000mAh = 5Ah$$

For safety and battery longevity, only 80% usable capacity is considered.

$$C_{usable} = 5 \times 0.8 = 4Ah$$

Thus, the practical usable battery capacity becomes:

$$4Ah$$

This usable capacity is considered for endurance calculations.

## Current Consumption

The estimated current consumption under different operating conditions is summarized below.

| Flight Condition          | Propulsion Current | Avionics Current | Total Current |
|---------------------------|--------------------|------------------|---------------|
| Hover Flight              | 13.6 A             | 2 A              | 15.6 A        |
| Normal Autonomous Mission | 18 A               | 2 A              | 20 A          |
| Aggressive Maneuvering    | 32 A               | 2 A              | 34 A          |
| Maximum Burst Load        | 45+ A              | 2 A              | 47+ A         |

The avionics current remains nearly constant because the Jetson Orin Nano, cameras, telemetry system, and sensors operate continuously throughout the mission.

## Endurance Estimation

The flight endurance is estimated using:

$$Endurance = \frac{Battery\ Capacity\ (Ah) \times 60}{Average\ Current\ Draw\ (A)}$$

### Case A : Hover Flight

$$I_{hover} = 15.6A$$

$$T_{hover} = \frac{4 \times 60}{15.6} \approx 15.4\ minutes$$

### Case B : Autonomous Mission Flight

$$I_{mission} = 20A$$

$$T_{mission} = \frac{4 \times 60}{20} \approx 12\ minutes$$

### Case C : Aggressive Flight Conditions

$$I_{aggressive} = 34A$$

$$T_{aggressive} = \frac{4 \times 60}{34} \approx 7.05\ minutes$$

## Expected Flight Time

Based on the propulsion calculations, avionics power budget, and usable battery capacity, the expected endurance of the UAV is summarized below.

| Flight Mode               | Estimated Flight Time |
|---------------------------|-----------------------|
| Hover Flight              | ~15 min               |
| Autonomous Mission Flight | ~12 min               |
| Aggressive Flight         | ~7 min                |

Thus, the UAV is expected to achieve approximately 10–15 minutes of practical autonomous flight endurance while continuously operating the onboard AI computing, sensing, telemetry, and navigation systems.

## 2.4 Material Selection

### Material Selection for Frame Design

A comparative analysis of different structural materials was performed before selecting the final material for the drone frame. The comparison was based on density, structural strength, manufacturability, and suitability for lightweight UAV applications.

Table 2.1: Comparison of Candidate Frame Materials

| Material         | Density      | Strength                  | Remarks                                                                                          |
|------------------|--------------|---------------------------|--------------------------------------------------------------------------------------------------|
| Carbon Fiber     | 1.5–1.6 g/cc | High tensile strength     | Carbon fiber is lightweight and strong but relatively expensive.                                 |
| Glass Fiber      | 2.4–2.6 g/cc | Moderate to high strength | Glass fiber is cheaper and strong but heavier than carbon fiber.                                 |
| PLA              | 1.24 g/cc    | Low strength              | PLA is inexpensive and easy to 3D print but has lower heat resistance and structural durability. |
| Aluminium Alloys | 2.7 g/cc     | Moderate to high strength | Aluminium alloys are strong and durable but significantly heavier than carbon fiber.             |

From the above comparison, carbon fiber was selected as the primary material for the drone frame due to its excellent strength-to-weight ratio, high stiffness, and superior vibration resistance, making it highly suitable for lightweight autonomous UAV applications.

### Landing Gear Material

Carbon fiber is also selected for the landing gear due to its very high strength-to-weight ratio and strong vibration resistance characteristics. The material is capable of withstanding landing impacts while maintaining low structural weight.

The bottom section of the landing gear is additionally covered with polystyrene padding to absorb impact forces during landing. This helps reduce shock transfer to the airframe and improves landing stability while protecting onboard electronics and payload components.

## 2.5 Subsystem Selection

### Communication System

The drone's communication architecture operates across two links.

The pilot's RC transmitter communicates with the onboard receiver over 2.4GHz using ELRS, delivering low-latency control inputs for manual override, with the receiver outputting CRSF signals directly to the flight controller.

A telemetry module connected to the FC via UART streams MAVLink packets over 2.4GHz to the ground station, providing real-time flight data including altitude, GPS, battery status, and arm state, while also allowing parameter tuning through Mission Planner or QGround-Control.

### Control and Navigation System

The control and navigation architecture follows a two-layer hierarchy. The inner control layer runs entirely on the CubePilot Orange flight controller, executing attitude control, rate control, and motor mixing at 400 Hz using the triple-redundant IMU data. This layer is responsible for stabilising the airframe against disturbances such as wind gusts and rotor vibration, and it operates independently of the companion computer at all times.

The outer navigation layer runs on the Jetson Orin Nano under ROS 2 Humble, implementing the mission state machine that determines where the drone should fly during each phase of the autonomous mission. The Jetson sends position setpoints and velocity commands to ArduCopter's GUIDED mode via MAVROS over MAVLink, while the inner loop on the Cube translates these into motor commands.

### Flight Controller

#### Flight Controller - CubePilot Orange

| Feature    | Specification                                |
|------------|----------------------------------------------|
| Processor  | STM32H757 Dual Core, 400 MHz                 |
| Memory     | 2 MB Flash, 1 MB RAM                         |
| Sensors    | ICM-42688, ICM-20948, ICM-20649, MS5611      |
| Interfaces | 14× PWM, 5× Serial, 2× CAN, 2× I2C           |
| Backup     | Dedicated processor, standalone power supply |

Table 2.2: Flight Controller Specifications

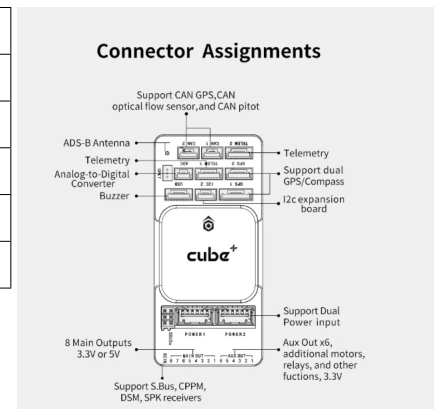


Figure 2.4: CubePilot Orange

## GPS Module

**Chipset:** U-blox M9N GNSS receiver with concurrent reception of 4 constellations (GPS, GLONASS, Galileo, BeiDou), maximising satellite lock in challenging environments.

**Processor:** Integrated STM32G4 microcontroller running the DroneCAN protocol stack.

**Protocol:** DroneCAN bus provides superior noise immunity, reliable differential signalling, and allows multiple peripherals to share a single bus without signal degradation.



Figure 2.5: Holybro DroneCAN M9N GPS

## Transmitter and Receiver



Figure 2.6: RadioMaster Pocket



Figure 2.7: RP1 ExpressLRS nano

The drone uses the RadioMaster Pocket transmitter paired with the RP1 ExpressLRS nano receiver, operating over the 2.4 GHz band. The Pocket runs EdgeTX firmware, features hall-effect gimbals, and supports 16 channels, with its integrated ELRS module providing up to 250 mW output and a 1 kHz link refresh rate. The ELRS protocol is selected for its robust, interference-resistant communication and sub-millisecond latency, ensuring responsive and reliable control.

## Companion Computer

### Companion Computer — NVIDIA Jetson Orin Nano 8GB

| Feature        | Specification                          |
|----------------|----------------------------------------|
| GPU            | Ampere, 1024 CUDA Cores, 32 Tensor Cor |
| CPU            | 6-core Arm Cortex                      |
| AI Performance | 40 TOPS                                |
| Memory         | 8 GB                                   |
| Power          | 7 W – 15 W                             |
| Communication  | MAVLink 2.0 over UART (921600 baud)    |
| OS             | ROS2 Humble with MAVROS                |

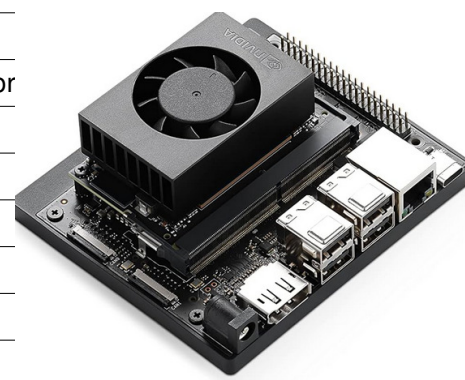


Table 2.3: NVIDIA Jetson Orin Nano 8GB Specifications

Figure 2.8: NVIDIA Jetson Orin Nano 8GB

The Jetson Orin Nano serves as the central decision-making unit for all autonomous operations in the mission. It communicates with the CubePilot Orange flight controller over

MAVLink 2.0 via UART, and concurrently runs five ROS2 nodes — the QR detection node, the vision node for red zone and banner detection, the corridor PID controller, the mission state machine, and the MAVROS telemetry bridge — without memory pressure or pipeline stalls.

## Power Management



Figure 2.9: BEC

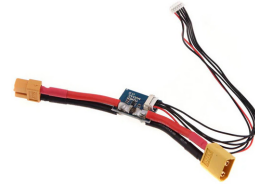


Figure 2.10: Power Module

The architecture employs a parallel power distribution strategy to isolate flight-critical components. A central LiPo feeds the Cube orange and ESCs via a Power Module, while a separate external BEC powers the Jetson Nano and other electronic components. This ensures reliability by decoupling sensitive flight logic from auxiliary electrical noise and voltage drops.

## Telemetry System

- **Frequency:** 2.4GHz
- **Range:** 15 km+ Line of Sight
- **Connection:** TELEM1 port on the Orange Cube
- **Role:** Provides a robust bi-directional MAVLink link between the drone and GCS, ensuring reliable monitoring of battery, altitude, and flight mode status at extended range.



Figure 2.11: LR2 4 2.4GHz LoRa Telemetry

## Camera System

### Camera — Arducam 16MP IMX519

- **Specs:** Sony IMX519 BSI, 1/2.534-inch, 16 MP, connected to the Jetson Orin Nano via MIPI CSI. 1920×1080 at 60 fps and 1280×720 at 120 fps for real-time processing.
- **Global Shutter:** Captures all pixels simultaneously, eliminating motion-induced distortion — critical for reliable QR code decoding up to 10 m.
- **Role:** Downward-facing; primary sensor for QR code detection, delivery zone identification, and aerial observation.

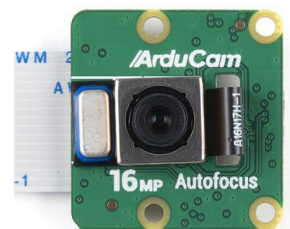


Figure 2.12: Arducam 16MP IMX519

## FPV System



Figure 2.13: VTX



Figure 2.14: RunCam Phoenix 2 JB Edition



Figure 2.15: Foxeer Pagoda PRO Antenna



Figure 2.16: Eachine goggles

**Camera:** RunCam Phoenix 2 JB Edition

**Transmitter:** SpeedyBee TX800

**Receiver:** Eachine EV800D

**Antenna:** Foxeer Pagoda Pro RHCP

The FPV system forms the pilot's primary sensory link during flight. The RunCam Phoenix 2 Joshua Bardwell Edition provides a sharp, wide-angle analog feed with high-sensitivity low-light performance and strong dynamic range across high-contrast environments. The SpeedyBee TX800 transmits this feed at up to 800mW over with selectable power levels and channels to minimize interference, radiated through the Foxeer Pagoda Pro a compact RHCP omnidirectional antenna with consistent coverage and multipath rejection. The Eachine EV800D goggles deliver the low-latency immersive view needed for precise manual flight, while its detachable monitor mounts on a tripod for the ground team to track flight paths and verify mission data simultaneously.

## Obstacle Detection Sensors



Figure 2.17: Depth Sensor

| Parameter        | Specification                                 |
|------------------|-----------------------------------------------|
| Depth Range      | 0.1 m – 20 m                                  |
| Depth Resolution | 1280×800 @ 30 fps / 848×480 @ 60 fps          |
| Field of View    | 86° H × 55° V                                 |
| IR Shutter       | Global shutter on IR channels                 |
| IR Illumination  | Active 850 nm projector with IR-Pass filter   |
| IMU              | 6-DoF gyroscope and accelerometer, 50–1000 Hz |
| Interface        | USB-C to Jetson Orin Nano via Orbbec SDK      |

Table 2.4: Orbbec Gemini 330 Stereo Camera Specifications

The Orbbec Gemini 330 is selected for its wide 0.1–20,m depth range and active IR illumination, which ensures reliable obstacle detection in outdoor sunlight conditions. The global shutter IR channels eliminate motion blur during flight, and the integrated IMU complements depth data for accurate real-time mapping. Its USB-C interface allows direct integration with the Jetson Orin Nano via the Orbbec SDK, making it well suited for onboard obstacle avoidance and corridor navigation.

## Range Finder

**Sensor:** Combined PMW3901 optical flow and ToF rangefinder module, measuring ground distance from 0.02 m to 8 m at 2% accuracy, rated to 70,000 Lux for outdoor use.

**FC Integration:** Connected via UART/MAVLink, feeding the PX4 EKF2 altitude channel for stable hold during low-altitude phases. Optical flow provides supplementary lateral velocity estimates below 3 m during final descent.



Figure 2.18: MTF-01P Optical Flow and Range Sensor

## 2.6 Stability Analysis of the Quadcopter

### Center of Gravity (CG) Estimation

Quadcopter stability relies heavily on the center of gravity (CG). Using CAD mass-properties analysis, the CG coordinates were determined as:

$$CG = (4.302, 8.632, 3.872) \text{ mm}$$

The CG lies near the frame's geometric center, ensuring balanced mass distribution. The x- and y-axis offsets are negligible relative to the 443.82 mm frame size.

The percentage offset along the x-axis is:

$$CG \text{ Offset}_x = \frac{4.302}{443.82} \times 100 = 0.97\%$$

Similarly, the offset along the y-axis is:

$$CG \text{ Offset}_y = \frac{8.632}{443.82} \times 100 = 1.94\%$$

Both offsets fall well within acceptable limits. This near-centered CG ensures minimal attitude correction is needed during hover, optimizing autonomous flight stability.

### Moment of Inertia Analysis

CAD-derived moments of inertia are:

$$I_{xx} = 1.793 \times 10^7 \text{ g mm}^2$$

$$I_{yy} = 2.021 \times 10^7 \text{ g mm}^2$$

$$I_{zz} = 2.152 \times 10^7 \text{ g mm}^2$$

The close values of  $I_{xx}$  and  $I_{yy}$  imply similar roll and pitch characteristics. The inertia variation is:

$$\Delta I = \frac{|I_{yy} - I_{xx}|}{I_{yy}} \times 100$$

$$\Delta I = \frac{|2.021 - 1.793|}{2.021} \times 100 = 11.3\%$$

A variation  $< 15\%$  indicates predictable and stable attitude response.

The higher yaw inertia ( $I_{zz} = 2.152 \times 10^7 \text{ g mm}^2$ ) prevents excessive yaw sensitivity and improves directional stability.

## Product of Inertia Analysis

The products of inertia are:

$$I_{xy} = 82774.019$$

$$I_{xz} = -3.733 \times 10^5$$

$$I_{yz} = -8.875 \times 10^5$$

These minimal values highlight very low inertial coupling between axes. Verification:

$$\frac{I_{yz}}{I_{zz}} \times 100 = \frac{8.875 \times 10^5}{2.152 \times 10^7} \times 100 = 4.12\%$$

Since cross-coupling remains  $< 5\%$ , the drone will exhibit minimal unwanted interaction between roll, pitch, and yaw, facilitating smoother autonomous control.

## Structural Stability Assessment

The symmetric X-frame configuration features equal motor spacing of 496.435 mm. This geometry guarantees balanced thrust and uniform torque distribution.

Centralizing the payload and battery minimizes destabilizing oscillations, ensuring reliable flight dynamics.

## Stability Conclusion

The designed quadcopter meets all primary stability requirements for autonomous missions. Key validation points include:

- A near-centered CG and balanced mass distribution.

- Acceptable rotational inertia symmetry with low cross-axis coupling.

- A fully symmetric rotor configuration.

The platform is mechanically sound and highly suitable for autonomous navigation, surveillance, and payload delivery.

## 2.7 Frame Design and Analysis

The custom carbon fiber quadcopter frame was designed for the AeroTHON 2026 competition to optimize subsystem integration, minimize structural weight, and enhance flexibility over commercial off-the-shelf alternatives.

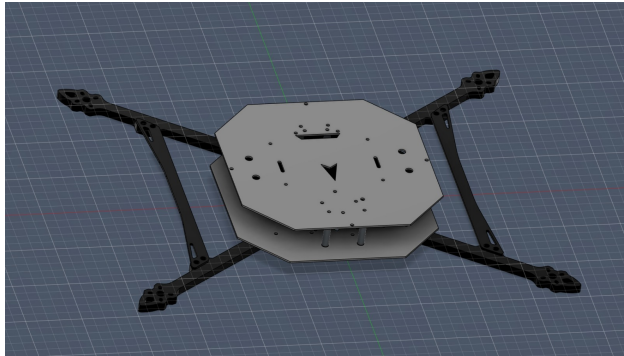


Figure 2.19: 3D CAD model

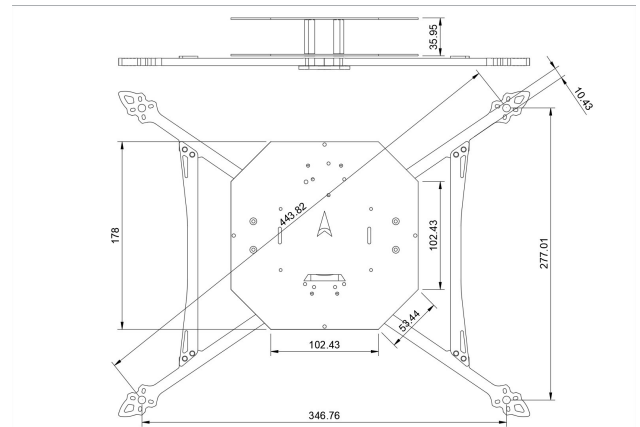


Figure 2.20: 2D dimensional drawing

The frame utilizes a symmetric X-configuration (inspired by the MARK4 10-inch) for proven stability and balanced thrust. The motor-to-motor diagonal is 443.82 mm, with an overall span of 346.76 mm.

A 102.43 mm  $\times$  102.43 mm central hub provides ample room for the flight controller, power distribution, autonomous navigation hardware, and battery.

By using custom 1.5 mm carbon fiber plates, unnecessary material and unused mounting points were eliminated, significantly reducing the frame's total mass.

Commercial frames in this size class were rejected because they either lacked payload space, exceeded Micro-UAS weight limits, or required extra mounting hardware.

The custom frame directly resolves these issues, enabling centralized component placement, an optimized center of gravity, reduced wiring complexity, and cleaner integration.

Carbon fiber construction dramatically improves the stiffness-to-weight ratio while damping structural vibrations—a critical requirement for accurate QR detection, corridor navigation, and precise autonomous payload drops.

Ultimately, this mission-specific design safely supports payload capabilities while remaining strictly under the 2 kg maximum take-off weight limit.

## 2.8 Computational Analysis

## Structural Analysis

We designed a custom 10-inch quadcopter frame with 5 mm thick T700 carbon fiber arms, offering superior stiffness-to-weight ratio and significantly raised natural resonant frequency. This rigidity minimizes low-frequency vibrations from larger propellers and provides a stable anchor point for the ducted fan assemblies and aerodynamic electronics bay. The frame's resistance to bending and twisting is critical for maintaining stable flight characteristics under the substantial leveraging forces generated by the 8-inch propeller system.

## Finite Element Analysis (FEA)

To validate structural integrity, we conducted static loading FEA on the CAD model with 3 mm mesh size. A thrust value of 28 N (100% maximum thrust of the Speedybee 2807-1050KV

motor) was applied at motor mounting points, with the central avionics bay as fixed support, simulating worst-case full-throttle conditions.

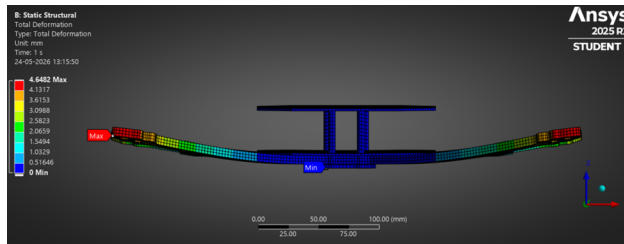


Figure 2.21: Total deformation distribution under 28 N maximum thrust loading showing 4.65 mm peak deflection at motor mounts

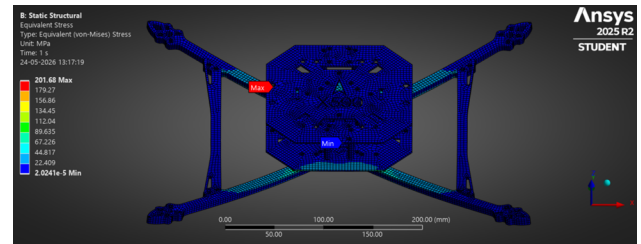


Figure 2.22: Equivalent von-Mises stress distribution showing maximum stress of 201.68 MPa at arm-center plate junctions

Results showed maximum total deformation of 4.65 mm at arm tips and maximum von-Mises stress of 201.68 MPa at arm-center plate junctions. With carbon fiber yield strengths ranging from 600-800 MPa, this indicates a substantial safety factor of approximately 3:1. Maximum elastic strain of 0.00375 mm/mm and peak strain energy of 0.476 mJ confirm the frame operates well within its elastic limit, ensuring no permanent deformation or structural fatigue over time.

## CFD Analysis

Aerodynamic analysis was conducted using Ansys Fluent to evaluate an 8-inch tri-blade propeller at 17,500 RPM (Speedybee 2807-1050KV, 6S configuration). A transient sliding mesh approach partitioned the geometry into stationary outer enclosure and rotating inner cylindrical cell zones with open interior mesh interface. Scalable Wall Functions managed extreme velocity gradients without solution divergence.

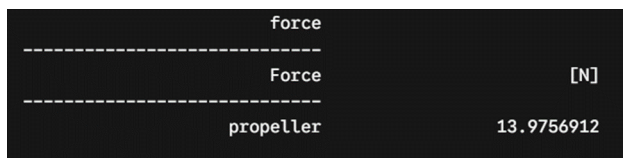


Figure 2.23: Thrust force monitor displaying calculated static thrust of 13.97 N at 17,500 RPM operational condition

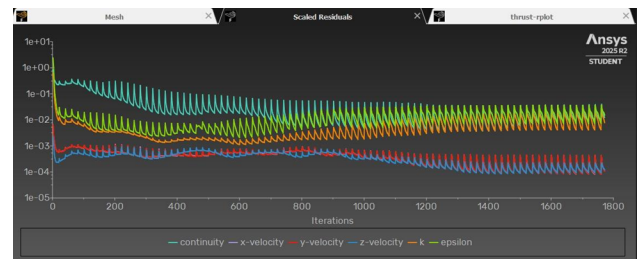


Figure 2.24: Scaled residuals plot confirming steady-state CFD solution convergence for the propeller simulation

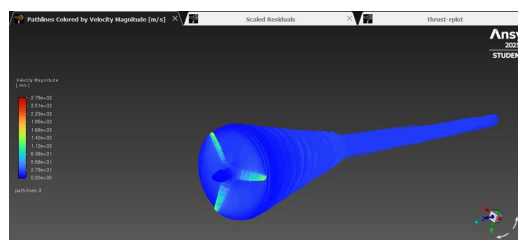


Figure 2.25: Velocity pathlines demonstrating stable helical wake formation and efficient air-flow through the rotating propeller domain

The solver yielded 13.97 N thrust at 17,500 RPM (approximately 50% throttle). Speedybee datasheet indicates 16-18 N at this RPM; the 3-4 N variance reflects differences between

computational static simulations and experimental dynamic testing (ground proximity effects, ambient turbulence, aeroelastic blade deformation). The simulation employed generic CAD geometry versus proprietary production airfoil profiles, providing a conservative baseline for aerodynamic integration.

## Modal Analysis

Modal analysis identified natural frequencies to ensure no coincidence with propulsion system operational frequencies. The first 12 natural frequencies are listed below.

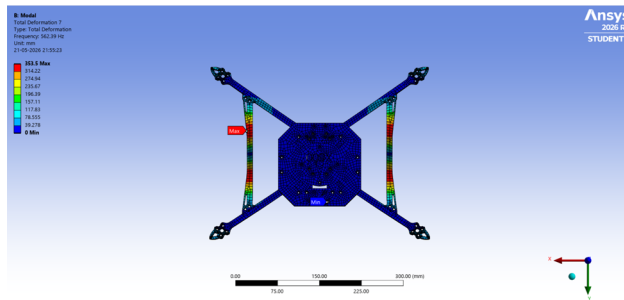


Figure 2.26: Twelfth mode shape at 562.39 Hz exhibiting complex torsional-bending coupling at motor mount locations

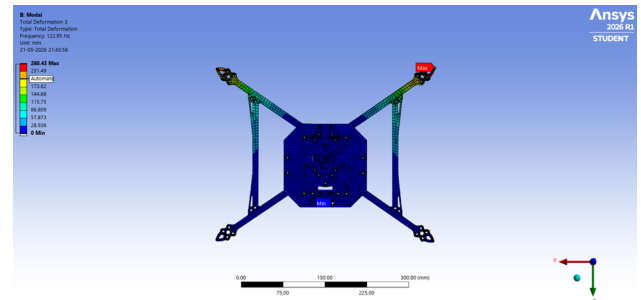


Figure 2.27: First mode shape at 122.95 Hz showing symmetric arm bending deformation pattern

Table 2.5: First 12 Natural Frequencies of the Drone Frame

| Mode | Frequency [Hz] |
|------|----------------|
| 1    | 122.95         |
| 2    | 127.41         |
| 3    | 147.22         |
| 4    | 158.03         |
| 5    | 161.49         |
| 6    | 224.50         |
| 7    | 227.45         |
| 8    | 279.08         |
| 9    | 398.65         |
| 10   | 476.40         |
| 11   | 536.80         |
| 12   | 562.39         |

The first natural frequency occurs at 122.95 Hz, providing wide frequency margin from motor operating range, successfully avoiding critical resonance and improving flight stability and structural durability.

## Harmonic Analysis

Harmonic analysis evaluated steady-state dynamic response from 0-500 Hz with 100 intervals and 0.02 damping ratio. A 28 N forcing load was applied at motor mounts with fixed support at the base plate.

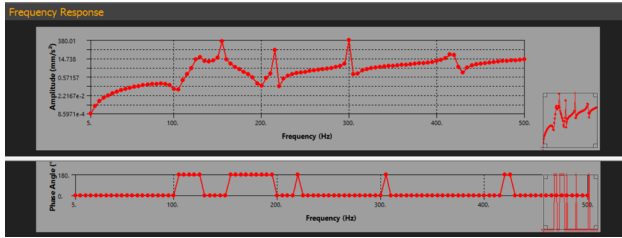


Figure 2.28: Acceleration frequency response plot showing peak amplitude of 380 mm/s<sup>2</sup> at 150 Hz resonance frequency

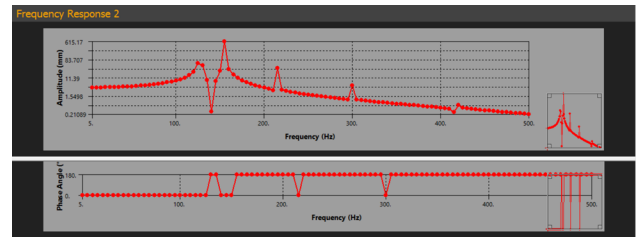


Figure 2.29: Displacement frequency response revealing maximum deformation amplitude of 615 mm at 150 Hz harmonic excitation

Prominent resonance peaks occur at 150 Hz, 220 Hz, 300 Hz, and 441 Hz, with the 150 Hz mode showing highest amplitude (615 mm). At 441 Hz, maximum deformation was 0.416 mm at arm tips. Within the operational range of 5-100 Hz, displacement amplitudes remain well-controlled, confirming structural integrity. The identified resonance frequencies lie outside typical propeller rotational frequencies (235-441 Hz for 14,100-26,460 RPM), providing adequate separation margin to avoid sustained resonant vibration during flight.

## 2.9 Payload Mechanism

The payload delivery system is built around a lightweight, dual-servo pulley architecture designed to lower the payload from a 5-meter hover while minimizing disturbance to flight stability.

It utilizes a single continuous high-strength string that spools from a primary MG90S 360° digital servo pulley, threads downward through the payload, and returns upward into the chassis to anchor onto a secondary 180° lock servo.

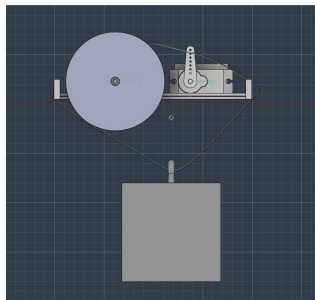


Figure 2.30: payload mechanism

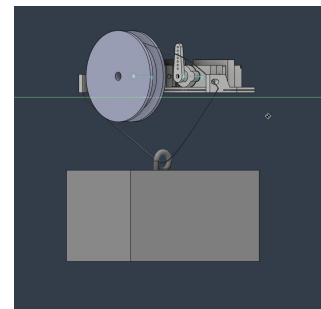


Figure 2.31: isometric view of payload mechanism

Upon receiving the drop command, the primary servo unspools the string at a carefully regulated speed, preventing sudden jerk forces from destabilizing the drone.

Once touchdown is detected, the lock servo rotates 90 degrees to release the loop, and the primary pulley instantly reverses to fully retract the string, ensuring no trailing lines remain exposed before the drone resumes forward flight.

## 2.10 Safety Mechanisms

**Geofencing:** Virtual boundaries restrict the drone to a predefined operational zone.

**Low Battery:** On low battery detection, the drone autonomously initiates landing, preventing mid-flight power loss and uncontrolled descent.

**Depth Camera Obstacle Avoidance:** A depth camera provides real-time spatial data to detect and avoid walls and obstacles, enabling safe navigation in confined environments.

**Bottom-Mounted Battery:** The LiPo battery is positioned at the bottom for low centre of gravity and quick manual power cutoff in emergencies.

**Isolated BEC with Insulated Wiring:** A dedicated, insulated BEC draws power directly from the battery (not the flight controller), isolating power domains and minimising short-circuit risk.

**Manual Override in Autonomous Flight:** If abnormal behaviour is detected during autonomous operation, the pilot can instantly reclaim control via the transmitter, overriding autonomous commands at any time.

## 2.11 Detailed Weight Breakdown and C.G. Analysis

| Category     | Component Description                                    | Qty | Unit Wt. (g) | Total Wt. (g)   |
|--------------|----------------------------------------------------------|-----|--------------|-----------------|
| Electronics  | Fullsend 6S 5000mAh Li-Ion Battery                       | 1   | 436.000      | 436.000         |
|              | UBEC 5V 5A                                               | 5   | 18.064       | 90.320          |
|              | SpeedyBee 2807 1050KV (6S) Motor                         | 4   | 47.000       | 188.000         |
|              | Orange Cube Standard Set (FC)                            | 1   | 105.023      | 105.023         |
|              | GEPRC 4-in-1 ESC                                         | 1   | 14.000       | 14.000          |
|              | Jetson Orin Nano (8GB)                                   | 1   | 171.000      | 171.000         |
|              | Holybro M9N GPS                                          | 1   | 37.000       | 37.000          |
|              | MTF-01 Optical Flow                                      | 1   | 4.500        | 4.500           |
|              | Orbbee Gemini 335                                        | 1   | 97.000       | 97.000          |
|              | Arducam 16 MP                                            | 1   | 14.412       | 14.412          |
|              | RunCam Nano 2                                            | 1   | 41.000       | 41.000          |
|              | SpeedyBee TX800 VTX                                      | 1   | 18.507       | 18.507          |
|              | RP4TD 2.4GHz ELRS Radio Master                           | 1   | 3.031        | 3.031           |
|              | Telemetry Casing                                         | 1   | 18.529       | 18.529          |
| Drone Parts  | Mark4 V2 Retained Base (4× Arms, Bottom Plate, Hardware) | 1   | 144.458      | 144.458         |
|              | X500 Frame Bottom Plate (2mm)                            | 1   | 57.850       | 57.850          |
|              | X500 Frame Top Plate (1.5mm)                             | 1   | 60.293       | 60.293          |
|              | Propellers                                               | 4   | 9.958        | 39.832          |
|              | Propeller Guards (Avg. 17.5g)                            | 4   | 17.500       | 70.000          |
|              | Landing Gear                                             | 1   | 102.000      | 102.000         |
|              | Dedicated Payload                                        | 1   | 100.000      | 100.000         |
|              | Payload Mech                                             | 1   | 36.930       | 36.930          |
|              | Enclosure                                                | 1   | 39.993       | 39.993          |
|              | Miscellaneous Components                                 | 1   | 30.000       | 30.000          |
| <b>TOTAL</b> | <b>Grand Total Takeoff Weight (AUW)</b>                  |     |              | <b>1919.698</b> |

The total AUW of **1919.698 g** confirms the UAS remains within the 2 kg MTOW constraint, with approximately 80 g of margin available for minor additions or component changes.

## 2.12 UAS sizing

Table 2.7: UAS Sizing Parameters

| Parameter           | Value     |
|---------------------|-----------|
| Wheelbase           | 443.871mm |
| Rotor Arm Length    | 227.172mm |
| Hub Diameter        | 9mm       |
| propeller pitch     | 4inch     |
| Propeller Diameter  | 8inch     |
| Ground Clearance    | 125.276mm |
| landing gear height | 212.776mm |

## 3. Methodology for Autonomous Operations

### 3.1 QR Code Detection and Identification

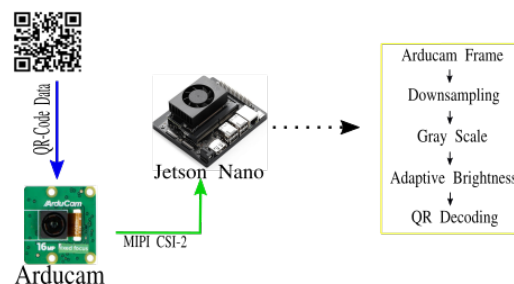


Figure 3.1: QR detection and Decoding

### QR Detection and Decoding Algorithm

QR code detection runs inside a dedicated processing node on the companion computer, using an open-source barcode decoding library to identify and read QR codes from each processed frame. Each camera frame is scanned for QR codes. When one is found, the encoded location identifier is extracted and passed to the next stage of the mission.

### Corridor Navigation

The drone detects the entry banner, aligns to the corridor centreline, and descends to 3 m. It then cruises at 0.8 m/s while two concurrent loops govern lateral position and obstacle response. Wall centering and forward motion run in parallel throughout transit. The drone avoids stopping unless the obstacle threshold forces it, keeping corridor transit time minimal. Altitude is managed independently using the LiDAR rangefinder, which is more reliable than barometric sensing at 3 m.

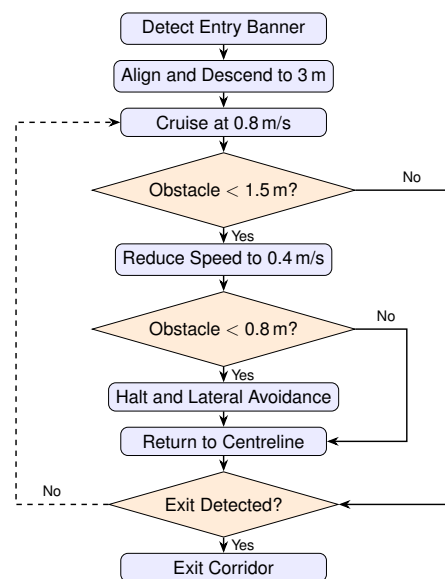


Figure 3.2: Corridor Navigation Decision Logic

## 3.2 Delivery Zone Identification

### Target Identification Method

After exiting the corridor, the drone climbs to 10 m and scans the ground below for QR codes. Every visible code is decoded and compared against the target ID collected at take-off. The first match is designated the delivery target, and its position in the camera frame is passed forward for coordinate estimation.

### Vision-based Localization

Once the target's ground coordinates are estimated, the drone descends to 5 m, continuously refining its position estimate as the QR code grows larger in frame. When the drone is within 15 cm of the target horizontally, it holds position and initiates payload release.

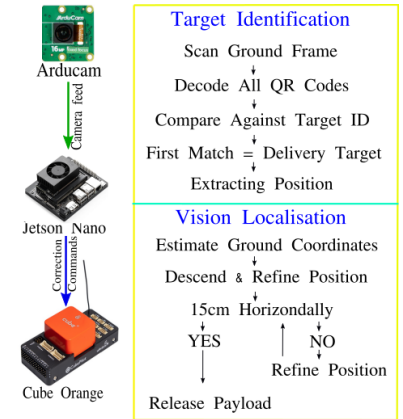


Figure 3.3: Delivery Zone Identification

## 3.3 Restricted Zone Avoidance

### Restricted Area Detection

The downward-facing camera continuously looks for red markings on the ground or on banners. Any red region large enough to represent a real boundary marker is flagged as a restricted zone.

### Avoidance Strategy

Avoidance operates at two levels. Pre-mapped restricted zones are enforced as hard boundaries in the flight controller, which the drone cannot cross. Additionally, any restricted zone detected live by the camera is treated like a physical obstacle, causing the path planner to route around it automatically.

### Safe Path Replanning

When a restricted zone appears ahead, the drone computes a curved path around it with at least 0.5 m clearance and follows that path as a series of updated position targets. Once safely past the zone, it returns to the original mission route.

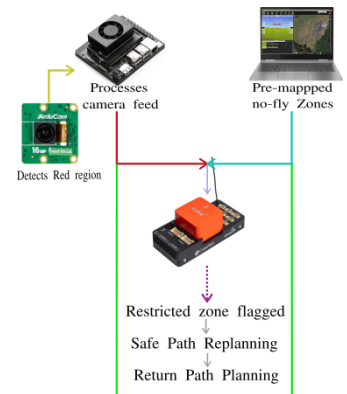


Figure 3.4: Restricted Zone Identification

## 3.4 Payload Delivery Methodology

The payload is suspended beneath the drone via a line routed through a pulley and secured by a hook. When the drone reaches the confirmed delivery position, the companion computer sends a rotation command to the servo. The servo rotates, releasing the hook, which allows the line to run through the pulley and lower the payload box smoothly to the ground.

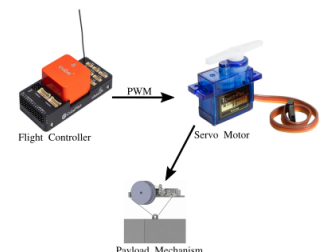


Figure 3.5: Payload Delivery Mechanism

## Autonomous Return and Landing

After payload delivery, the drone searches for the exit, navigates through the exit corridor avoiding obstacles, and executes the return and landing sequence illustrated in Figure 3.6.

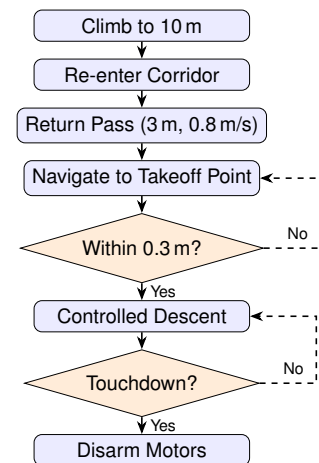


Figure 3.6: Return and Landing Logic

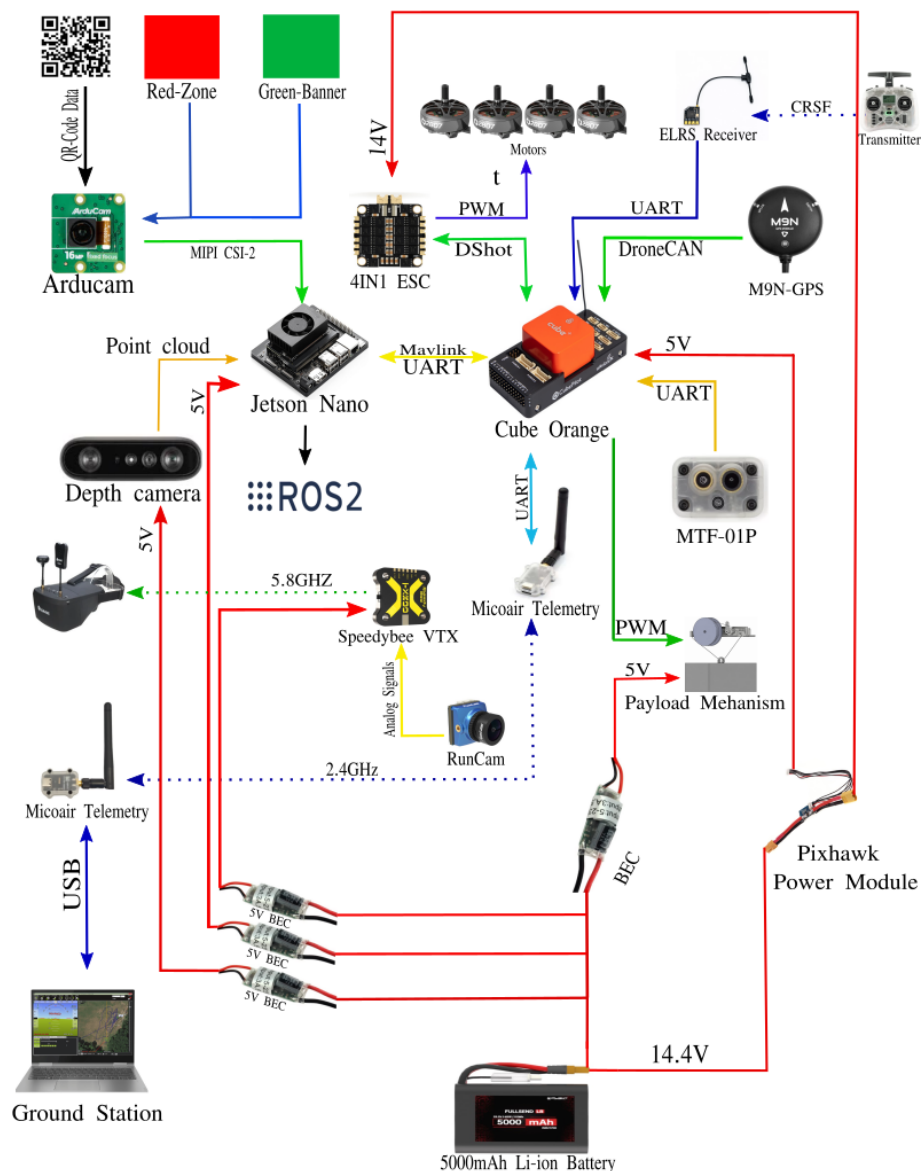


Figure 3.7: Stem pipeline

## 4. Final UAS Specifications and Bill of Materials

| Sl. No | Component                           | Qty | Cost ( ) | Remarks                              |
|--------|-------------------------------------|-----|----------|--------------------------------------|
| 1      | CubePilot Orange Flight Controller  | 1   | 36000    | Triple redundant IMU autopilot       |
| 2      | Holybro DroneCAN M9N GPS            | 1   | 7999     | DroneCAN GNSS module                 |
| 3      | RadioMaster Pocket ELRS Transmitter | 1   | 7500     | 16-channel RC transmitter            |
| 4      | RP1 ExpressLRS Nano Receiver        | 1   | 2300     | 2.4GHz ELRS receiver                 |
| 5      | NVIDIA Jetson Orin Nano 8GB         | 1   | 55000    | Companion computer for AI processing |
| 6      | BEC Module                          | 6   | 1500     | External regulated power supply      |
| 7      | Pixhawk Power Module                | 1   | 2500     | Power monitoring and regulation      |
| 8      | MicoAir LR900 Telemetry System      | 1   | 8500     | Long range LoRa telemetry            |
| 9      | MTF-01p LiDAR                       | 1   | 3500     | Altitude sensing LiDAR               |
| 10     | Arducam 16MP IMX519 Camera          | 1   | 4500     | QR detection and vision tasks        |
| 11     | RunCam Nano 2 FPV Camera            | 1   | 2700     | Low latency FPV camera               |
| 12     | SpeedyBee TX800 VTX                 | 1   | 3200     | 5.8GHz video transmitter             |
| 13     | Eachine EV800D FPV Goggles          | 1   | 10500    | FPV ground monitoring system         |
| 14     | Orbbec Gemini 330 Depth Camera      | 1   | 27000    | Stereo depth sensing camera          |
| 15     | MTF-01P Optical Flow Sensor         | 1   | 4500     | Optical flow and rangefinder         |
| 16     | Carbon Fibre Plate                  | 1   | 6000     | Base                                 |
| 17     | Frame                               | 1   | 7000     | Custom Frame                         |
| 18     | 3D Printing                         | 1   | 1500     | Approx.                              |
| 19     | DroneCAN Cable Set                  | 2   | 800      | CAN communication cables             |
| 20     | Fullsend 6S 5000mAh Li-Ion Battery  | 1   | 6400     | Battery                              |
| 21     | SpeedyBee 2807 1050KV Motor         | 4   | 11596    | 2,899/pc                             |
| 22     | Gemfan 8040 Propellers              | 2   | 658      | 329/pc                               |
| 23     | GEP-BLS50A-4IN1 ESC                 | 1   | 3795     | ESC                                  |
| 24     | Prop Guard                          | 4   | 899      | Custom designed                      |

# 5. Innovation and Conclusion

## 5.1 Innovation & System Optimization

To meet the rigorous operational requirements of the target mission profile, the vehicle incorporates two major custom-engineered optimizations: a minimalist payload architecture and active vibration damping.

### Ultra-Lightweight, Integrated Payload Architecture

Engineered to execute precise deployments from a 5-meter hover, the 37-gram payload mechanism is aggressively optimized for weight reduction and spatial compactness.

Instead of a bulky winch with heavy drop-hooks, it employs a continuous-loop routing system where a single string drops from the chassis, loops through the payload, and anchors back inside the drone.

This perfectly centralizes the center of gravity while eliminating swinging and tangling.

Upon ground contact, an automated lock servo releases one end of the loop while the primary pulley fully retracts the string, completely neutralizing drag and snag hazards before forward flight resumes.

### Structural and Acoustic Vibration Damping

Operating large 8-inch propellers generates significant low-frequency resonance that can rapidly degrade IMU and vision sensor accuracy. To mitigate this, the airframe employs a friction dampening architecture rather than relying exclusively on material stiffness. The carbon fiber rotor arms are structurally sandwiched tightly between the upper and lower central plates. Under high dynamic loads and motor vibrations, the micro-displacements at these overlapping joint interfaces generate Coulomb friction. This friction actively dissipates vibrational kinetic energy, effectively isolating the central avionics bay from high-amplitude resonant frequencies.

## 5.2 Conclusion

The developed Unmanned Aerial System (UAS) is a custom 10-inch symmetric quadcopter explicitly engineered for stability, autonomous processing, and precision payload delivery.

Utilizing a highly rigid T700 carbon fiber chassis, the drone safely houses a heavy AI vision stack—including a Jetson Orin Nano, Orbbec Gemini 330, and Arducam 16MP—while maintaining a highly efficient All-Up Weight of 1919.7 g.

Key achievements include a reliable 12–15-minute flight endurance, decoupled structural resonance (122.95 Hz) to protect sensor fidelity, and a robust 50 Hz reactive navigation loop for GPS-denied corridor transit and restricted-zone avoidance.

Furthermore, the custom 37-gram dual-servo pulley mechanism successfully enables perfectly balanced, swing-free payload descents from a 5-meter hover, followed by fully automated line retraction.

# A. Appendix

## A.1 Datasheets

### Battery Datasheet

#### Specifications

Capacity: 5000mAh

Voltage: 22.2V

Cell Type: EVE INR21700-50PL

Battery Capacity: 111Wh

Cell combination: 6S1P

Size: 43\*63\*75mm

Weight: 436(±5)g

Discharge rate: 25C

Battery type: Lithium-ion

Connector: XT60H

Discharge wire plug and length: XT60H-F(AMASS)-130(±8) mm

Charging cable plug and length: JST-XH2.54-7P-45mm

Max Voltage (Fully Charged): 25.2V (4.2V per cell)

Figure A.1: Specification of Battery used

### 2D Drawing of Motor

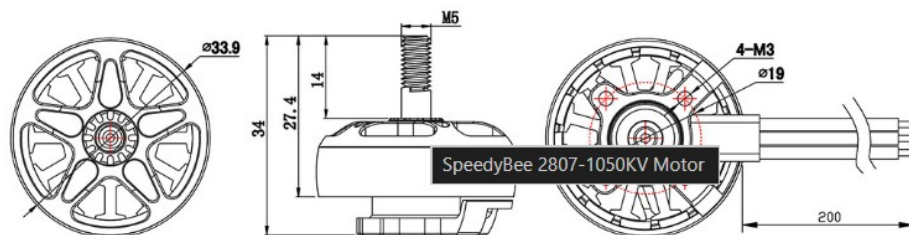


Figure A.2: 2D Drawing of Motor

## Datasheet of motor

| Motor Model | Voltage | Propeller Size                     | Current | Thrust | Power   | Efficiency | Rotational Speed |
|-------------|---------|------------------------------------|---------|--------|---------|------------|------------------|
| 2807-1050KV | 25.2    | Gemfan8040<br>3-blade<br>propeller | 1.8     | 300    | 45.36   | 6.61       | 6000             |
|             | 25.2    |                                    | 3.4     | 500    | 85.68   | 5.84       | 7860             |
|             | 25.2    |                                    | 5.3     | 700    | 133.56  | 5.24       | 9220             |
|             | 25.2    |                                    | 7.7     | 900    | 194.04  | 4.64       | 10300            |
|             | 25.2    |                                    | 10.3    | 1100   | 259.56  | 4.24       | 11300            |
|             | 25.2    |                                    | 47.2    | 2870   | 1189.44 | 2.41       | Max              |

SpeedyBee 2807-1050KV Motor

Figure A.3: Datasheet of motor

## ESC datasheet

### Specifications

- Input Voltage: 3-6S Lipo
- Galvanometer: Support
- Continuous Current: 50A
- Burst Current: 55A (5s)
- Support Protocol: Oneshot125/Oneshot42/Mulishot modes
- Dshot 150/300/600
- Size: 42.9×39.7mm, hole30.5×30.5mm,  $\phi$ 4mm hole becomes  $\phi$ 3mm after using the silicone shock-absorbing circle.
- Weight: 13.7g
- Target: G\_H\_30

Figure A.4: Specification of ESC used

## A.2 Full Assembly View

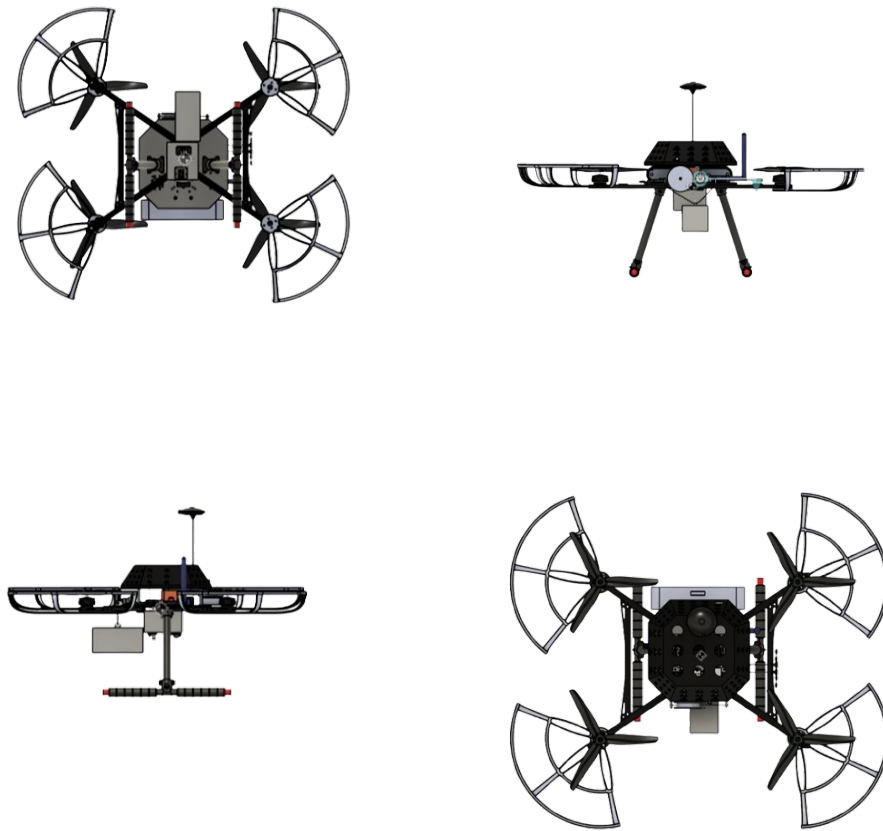


Figure A.5: Different views of Drone

A.3 Plansheet

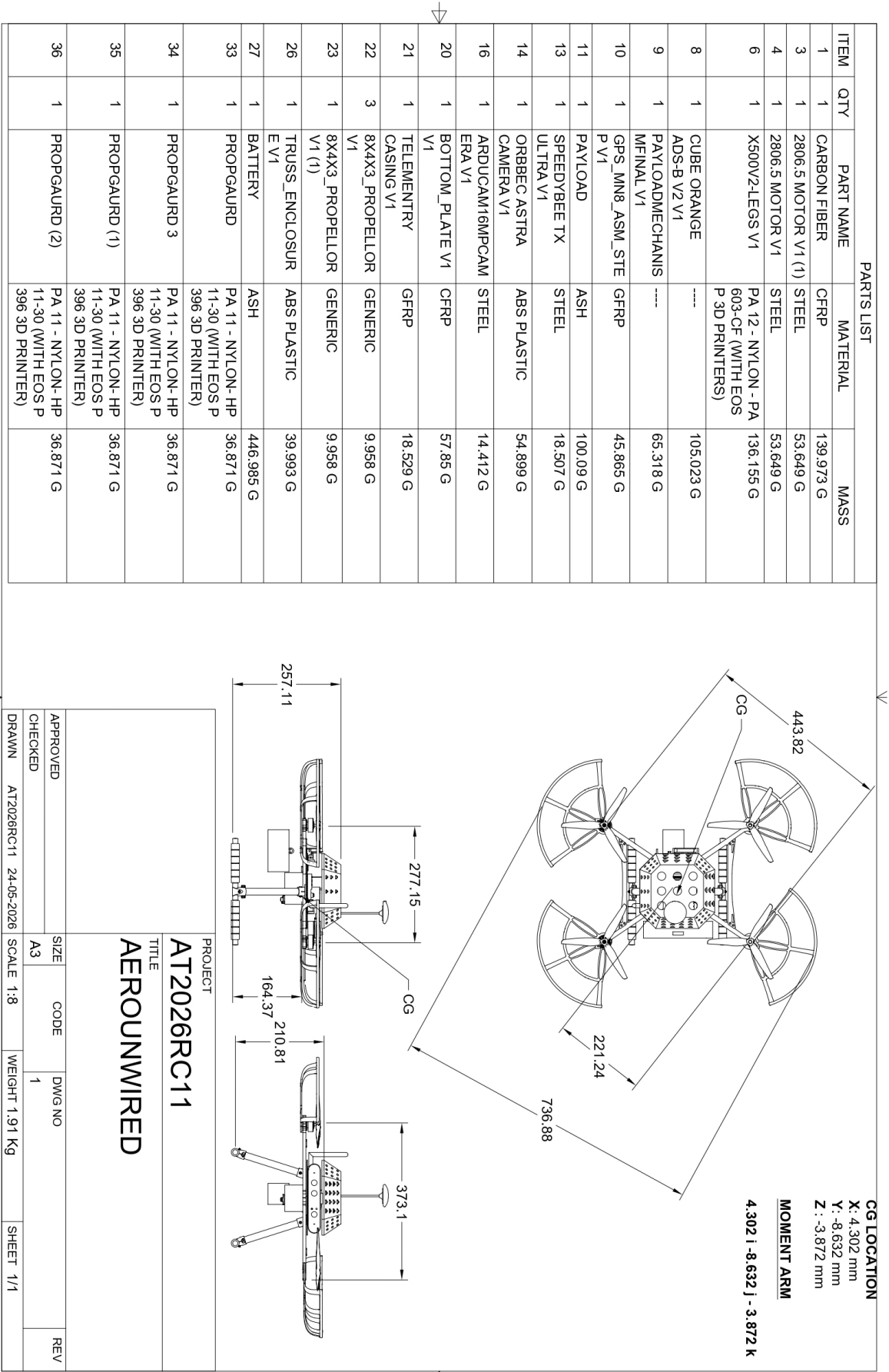


Figure A.6: Plansheet

A.4 Part Drawing

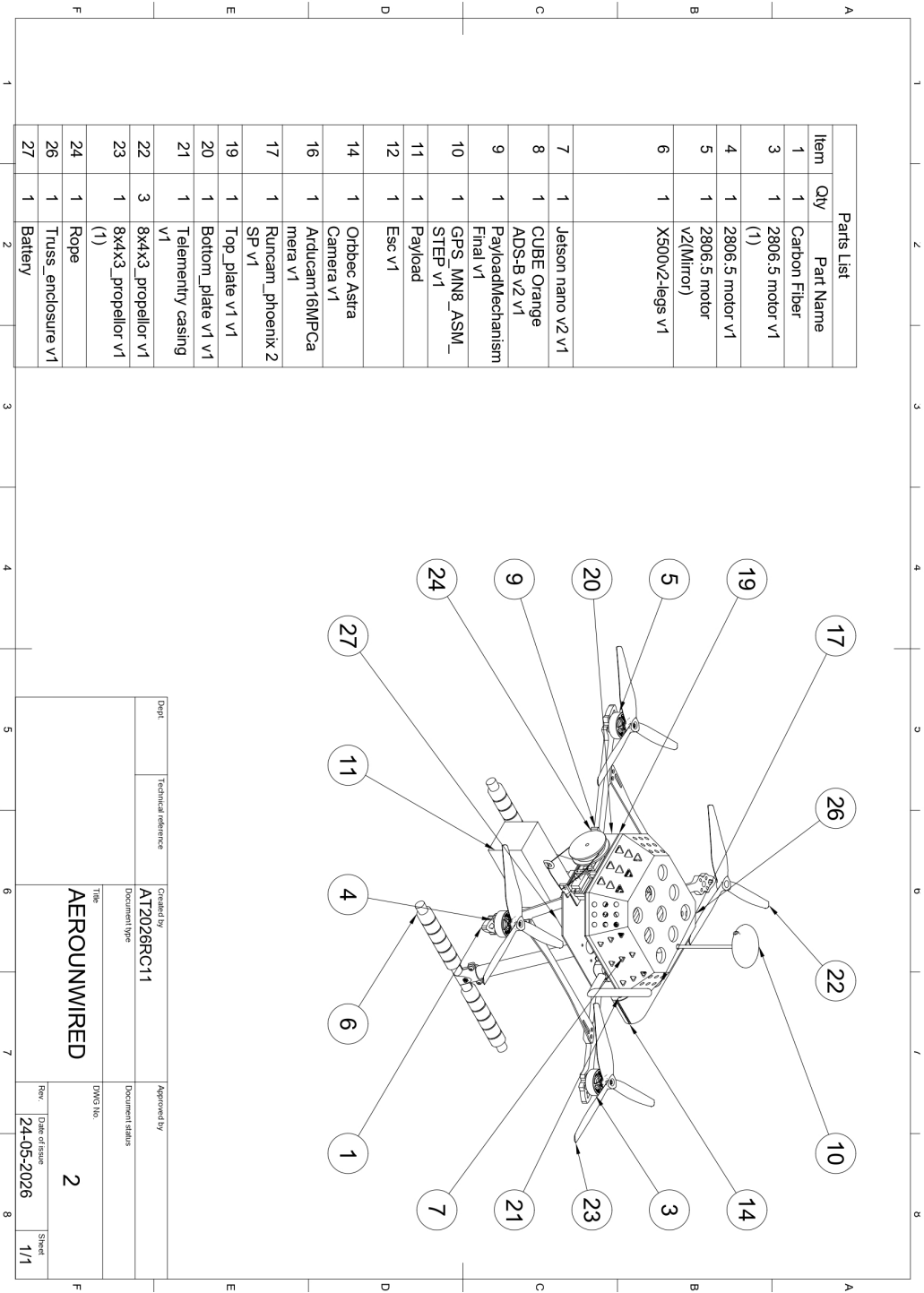


Figure A.7: Parts Drawing

# References

1. PX4 Development Team, *PX4 Obstacle Avoidance*. Available at: [\[Link\]](#)
2. Orbbec, *Orbbec Gemini 330 Series Supported Hosts and System Requirements (GMSL)*. Available at: [\[Link\]](#)
3. ExpressLRS, *ExpressLRS Serial Protocols*. Available at: [\[Link\]](#)
4. PX4 Discussion Forum, *Autonomous Mission using Servo Motors*. Available at: [\[Link\]](#)
5. ArduPilot, *Optical Flow Sensor Interference – MTF-01*. Available at: [\[Link\]](#)
6. ArduPilot, *Geofencing*. Available at: [\[Link\]](#)
7. ArduPilot, *Battery Fail Safe*. Available at: [\[Link\]](#)
8. Academia.edu, *Propeller CFD Analysis*. Available at: [\[Link\]](#)
9. SimScale, *Finite Element Analysis (FEA) Research*. Available at: [\[Link\]](#)
10. GrabCAD, *Drone CAD Files Library*. Available at: [\[Link\]](#)
11. Google Scholar, *Carbon Fiber Drone Frame Analysis*. Available at: [\[Link\]](#)
12. IEEE Xplore, *Quadcopter Research*. Available at: [\[Link\]](#)
13. Small Unmanned Aircraft: Theory and Practice — Reference for quadcopter dynamics, stability analysis, and UAV flight behavior.
14. Principles of Helicopter Aerodynamics — Reference for rotor aerodynamics and multi-rotor aerodynamic analysis.
15. ANSYS Modal — Used for modal and vibration analysis of the quadcopter frame to study natural frequencies and structural resonance.